

Polynomials – 7

Interpolation and finite differences

8 сентября 2026 года

Theoretical minimum.

Theorem 1 (Lagrange). Through any $n+1$ points with distinct abscissas there passes exactly one polynomial of degree at most n , namely $L(x) = \sum_i y_i \prod_{j \neq i} \frac{x-x_j}{x_i-x_j}$. Uniqueness is the statement that the Vandermonde determinant $\prod_{i < j} (x_j - x_i)$ does not vanish.

Theorem 2 (finite differences). For $\Delta f(x) = f(x+1) - f(x)$ the operator Δ lowers the degree by one and $\Delta^n f = n! a_n$ for $\deg f = n$. Hence $\sum_{k=0}^n (-1)^k \binom{n}{k} f(k) = 0$ whenever $\deg f < n$, and equals $(-1)^n n! a_n$ when $\deg f = n$.

Fact 1 (Newton's form). Every polynomial is uniquely $f(x) = \sum_k \Delta^k f(0) \binom{x}{k}$; it takes integer values at all integers exactly when all $\Delta^k f(0)$ are integers, and then it is an integer combination of binomial coefficients.

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1. Prove Theorem 1 and Theorem 2, and prove the identity $\sum_{k=0}^n (-1)^k \binom{n}{k} P(k) = 0$ for every polynomial P of degree less than n .
 2. (Putnam 2019) Let F_m be the Fibonacci numbers and let p be the polynomial of degree 1008 with $p(2n+1) = F_{2n+1}$ for $n = 0, 1, \dots, 1008$. Find integers j and k with $p(2019) = F_j - F_k$.
 3. (VJIMC 2011)
 - a) Is there a real polynomial P with $P\left(\frac{1}{k}\right) = k + \frac{2}{k}$ for all positive integers k ?
 - b) Is there a real polynomial P with $P\left(\frac{1}{k}\right) = \frac{1}{2k+1}$ for all positive integers k ?

4. (Putnam 1986) Let $a_1, \dots, a_n$ be real numbers and $b_1, \dots, b_n$ distinct positive integers, and suppose a polynomial f satisfies $(1-x)^n f(x) = 1 + \sum_{i=1}^n a_i x^{b_i}$. Find a closed expression for $f(1)$ in terms of $b_1, \dots, b_n$ and n .
5. (Putnam 2002) Let k be a fixed positive integer. The n -th derivative of $\frac{1}{x^k-1}$ has the form $\frac{P_n(x)}{(x^k-1)^{n+1}}$ with P_n a polynomial. Find $P_n(1)$.